

Title: Initial GSM-R testing at Stoke ref. lab (NR)**Test record No:** PS/NR-22-003/TR004

Purpose of Test

To make a site visit to Network Rail's (NR) reference lab at Stoke-on-Trent, and perform a series of tests to give confidence that we will be able to successfully utilise our preferred GSM-R modem (the TrioRail TRM-5) in the DMWS IDR Core Unit. We need to do this at the Stoke reference lab, because GSM-R GPRS coverage is generally only available at NR premises/tracksides, and needs specialist SIM cards, which only NR can configure and supply.

Objectives of the tests:

- 1) To test the ability of PSL's TrioRail TRM-5 GSM-R modem, to 'camp on' to (or connect to, and register with) NR's GSM-R network.
- 2) Having successfully connected, to set up a GPRS data connection between the TRM-5 and an ethernet device, via the GSM-R network.
- 3) To verify that the GPRS connection can be established without having to supply APN details.
- 4) Using the GPRS data connection to send/receive TCP and UDP to/from a PSL-supplied TCP/UDP server, which will be connected to a spare ethernet port in the reference lab.
- 5) Show we can read the ICCID from the SIM card electronically
- 6) Show we can read the IP address assigned to the GPRS terminal

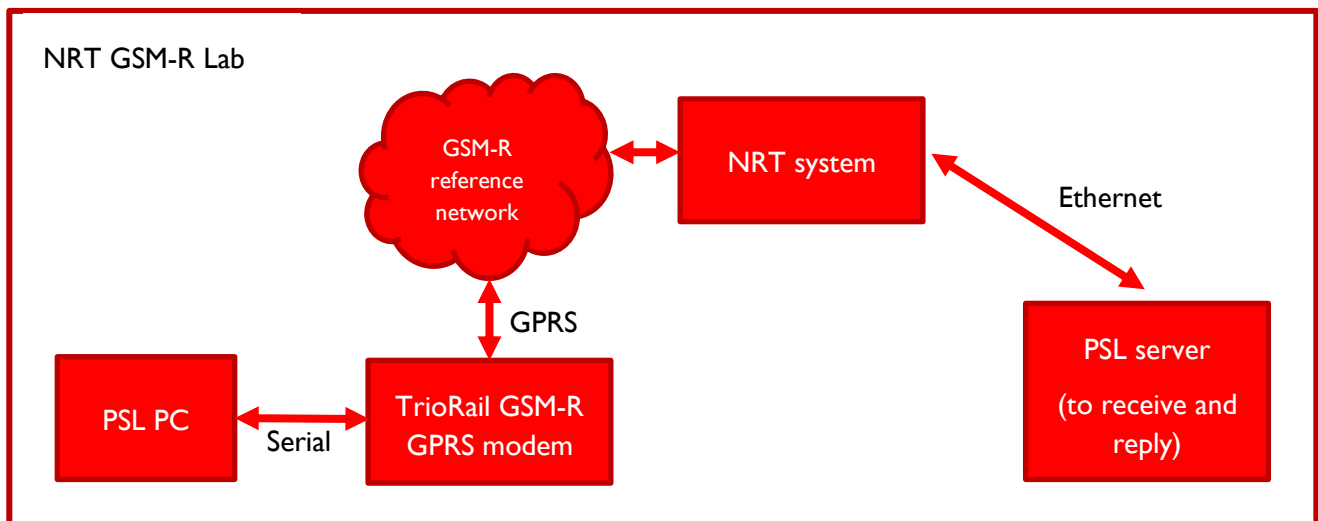
Notes:

PSL will require a correctly configured GSM-R mini-SIM card (~25mm x 15mm - we believe this is the size normally used by NR) to be supplied by NR.

The PSL-supplied server will need to be allocated a dynamic or static IP address by NR, and the allocated IP address will need to be communicated to PSL by NR. Suitable port numbers for TCP and UDP connections will need to be open/accessible through any firewalls/gateways on NR's infrastructure between the ethernet connection and GPRS connection.

Test method to be used

Note this is largely an informal test to build confidence.



'PSL PC' shall be capable of controlling the TrioRail module.

'PSL server' will be capable of accepting TCP connections (act as TCP server) on port 50000 and responding to TCP messages with another TCP message. It shall also respond to UDP messages on port 50001 by replying with an UDP packet on port 50001.

PSL barcode numbers of equipment used, and details of the software used on PSL PC/PSL server shall be recorded.

During the following testing, Wireshark software should be run on the PSL server in order to record network activity and all data on the PSL PC/TrioRail module shall be logged.

- 1) Setup a static IP address on PSL server as required. Connect via ethernet to NRT's systems.
- 2) Insert NR SIM into TrioRail module.
- 3) Power on TrioRail module, PSL PC and PSL server.
(It is assumed that NRT systems are already configured for the test, powered on, and ready.)
- 4) Use PSL PC to attempt connection to NR GSM-R GPRS network.
- 5) Use PSL PC to repeatably send UDP packets to port 50001 of the IP address of the PSL server.
- 6) Check that UDP packets are received on the PSL server.
- 7) Check that responses are received on the PSL PC/TrioRail module.
- 8) Stop sending UDP data on the PSL PC/TrioRail module.
- 9) Use PSL PC/TrioRail module to set up a TCP connection to port 50000 of the IP address of the PSL server.
- 10) Use PSL PC/TrioRail module to send TCP data on the established connection.
- 11) Check that the expected TCP data is received at the PSL server.
- 12) Check that responses are received on the PSL PC/TrioRail module via TCP.
- 13) Use PSL PC/TrioRail module to close TCP connection – record the result.
- 14) Use PSL PC/TrioRail module to query the TrioRail module's assigned IP address – record the result.
- 15) Use PSL PC/TrioRail module to query the SIM card ICCID – record the result.

Measures of success/failure

A connection to PSL server can be made via GSM-R GPRS, and UDP and TCP data transferred in both directions.

Test Defined by:	X	Test Accepted by:	X
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ResultsTest Equipment and hardware/software versions

Item	Serial number/Version

Conclusion

Date of test: Click or tap to enter a date.			
Test Performed by:	X	Test Checked by:	X